

PEOPLEFLOW

A Modern Enterprise HRMS & Leave Management Platform

DEVELOPED BY:

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INTERNSHIP HOST & MENTOR:

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Project Overview & Motivation

THE PROBLEM

- **Administrative Friction:** Traditional systems require heavy manual overhead for payroll, attendance, and leave management.
- **Scattered Records:** Employee profiles, department directories, and company documents exist in silos (spreadsheets, emails).
- **Unmanaged Assets:** No structured logs for IT asset assignments, causing lost inventory and delayed handovers.
- **Lack of Transparency:** Approvals are delayed due to offline request pipelines, hurting employee satisfaction.

THE PEOPLEFLOW SOLUTION

- **Unified Dashboard:** All operations (Attendance, Leaves, Assets, Analytics) consolidated under one portal.
- **Automated Tracking:** One-click digital check-in/check-out with real-time logging and analytics.
- **Streamlined Workflows:** Multi-stage digital leave requests and approval pipelines for managers and HR.
- **Audited Environment:** Full audit logging for actions (logins, updates, role changes) to ensure security.

System Architecture & Tech Stack

FRONTEND CLIENT

- **React.js & TS:** Typed components for highly scalable code structure.
- **Tailwind CSS:** Fully custom responsive design utilities.
- **Vite:** Next-gen lightning-fast frontend compilation.
- **Framer Motion:** Micro-interactions and fluid animations.

BACKEND SERVICES

- **Node.js & Express:** Asynchronous, fast REST API controllers.
- **Security & Auth:** JWT-based sessions, Bcrypt hashing, Helmet protection.
- **Middleware:** Role validation, rate-limiting, and error-handling filters.
- **Nodemailer:** SMTP integration for emails.

DATABASE & CLOUD

- **PostgreSQL:** Robust, relational schema holding transactions and profiles.
- **Render:** Production host for Node backend server.
- **Vercel:** Production host for static built frontend application.
- **Automatic Migrations:** Database schemas sync automatically on boot.

Core Product Features

EMPLOYEE DIRECTORY & PROFILES

- **Comprehensive Profiles:** Work history, salaries, document store, managers.
- **Dynamic Search & Filters:** Query by name, skills, role, or department.

DAILY ATTENDANCE MANAGEMENT

- **Check-in / Check-out:** Single-click digital registry.
- **Analytics Dashboards:** Visual metrics of monthly/weekly check-ins.

LEAVE WORKFLOWS & APPROVALS

- **Leave Balance Tracking:** Tracks allocated, used, and pending leaves.
- **Multi-level Approvals:** Routed dynamically to managers and HR.

AUDIT ENGINE & ROLE-BASED ACCESS

- **Granular Permissions:** Super Admin, HR, Manager, and Employee tiers.
- **Action Log Audits:** Secure trails logging logins, actions, and IP addresses.

Database Schema & Relations

PRIMARY DATA ENTITIES

- **users:** Credentials, status, locked status, and registration info.
- **employees:** Demographics, contact info, job level, salary, manager_id.
- **departments:** Stores department details and maps the head manager.
- **leave_requests:** Requested ranges, total days, approvals, and status.
- **assets:** Company inventory with serial codes and current assignments.

RELATIONAL ARCHITECTURE

- **1-to-1 Mapping:** users ↔ employees linked by user_id to isolate auth details from profile metadata.
- **Self-Referencing Manager Link:** employees.manager_id links to employees.id to resolve hierarchical approvals.
- **Junction Tables:** user_roles and role_permissions resolve multi-role assignments cleanly.
- **Integrity Rules:** ON DELETE CASCADE on tokens/logs prevents database orphan records.

Key Technical Accomplishments & Fixes

BACKEND & INFRASTRUCTURE

- **Cloud PostgreSQL Compatibility:** Upgraded pg database pool config to parse full DATABASE_URL connection strings.
- **SSL Configurations:** Integrated SSL (rejectUnauthorized: false) for cloud database providers.
- **Startup Migrations:** Refactored server.js to run migrations programmatically on startup, ensuring auto-creation of tables on Render.
- **Fault-tolerant Emails:** Made email verification optional and wrapped sendMail in try-catch to prevent mail server down-time from blocking registrations.

FRONTEND & PROXY LAYERS

- **Vercel Rewrite Proxy:** Created vercel.json rewrites for /api/* and /uploads/* pointing to Render.
- **CORS Fixes:** Resolved CORS blockages transparently via Vercel proxying and unified allowed headers.
- **React Router Fallback:** Configured Vercel's index.html fallback rewrite to prevent 404 errors when reloading sub-routes.
- **Flexible Client Base URL:** Modified Axios api.ts client to check for VITE_API_URL, matching localhost or cloud hosts dynamically.

Internship Learnings & Conclusion

TECHNICAL ACQUISITIONS

- **State Management:** Hands-on experience in React context and authentication stores.
- **Database Transactions:** Mastered SQL scripts, transaction commits, rollbacks, and indexes.
- **System Deployments:** Configured production builds, SSL databases, and API reverse proxying.
- **Auditing & RBAC:** Designed structured role authentication and security event auditing.

CONCLUSION & FUTURE ROADMAP

- **Industry Alignment:** Working on PeopleFlow during this internship at I-Soft Zone under Pranay Gupta provided exposure to enterprise coding standards.
- **Payroll Integration:** Plan to implement salary structures, tax deductions, and download pay slips.
- **AI Shift Planner:** Integrate models to forecast employee schedules and suggest optimization.
- **Thank You:** Deep gratitude to SUAS Indore and I-Soft Zone for this learning opportunity.